

Regression Statistics — Line-by-Line Calculation

Advertising Spend (X) → Sales (Y)

Advertising Spend (X)	Sales (Y)
10	105
12	112
15	120
18	128
20	135
22	141
25	150
28	158
30	165
32	171
35	180
38	188
40	195
42	202
45	212

Step 1 — Basic totals

Number of observations: $n = 15$

$$\Sigma X = 412$$

$$\Sigma Y = 2362$$

$$\Sigma X^2 = 13068$$

$$\Sigma Y^2 = 387906$$

$$\Sigma XY = 70164$$

$$\text{Mean } X = \Sigma X / n = 412 / 15 = 27.466667$$

$$\text{Mean } Y = \Sigma Y / n = 2362 / 15 = 157.466667$$

Step 2 — Multiple R

$$\text{Formula: } r = [n\Sigma XY - (\Sigma X)(\Sigma Y)] / \sqrt{[n\Sigma X^2 - (\Sigma X)^2][n\Sigma Y^2 - (\Sigma Y)^2]}$$

$$\text{Numerator} = 15(70164) - (412)(2362) = 79316$$

$$\text{First denominator part} = 15(13068) - (412)^2 = 26276$$

$$\text{Second denominator part} = 15(387906) - (2362)^2 = 239546$$

$$\text{Denominator} = \sqrt{(26276 \times 239546)} \approx 79336.691991$$

$$\text{Multiple R} = 79316 / 79336.691991 \approx 0.999739188$$

$$\text{Multiple R} = 0.999739188$$

Step 3 — R Square

Formula: $R^2 = (\text{Multiple R})^2$

$$R^2 = (0.999739188)^2 = 0.999478443$$

R Square = 0.999478443 ($\approx 99.9478\%$)

Step 4 — Adjusted R Square

Formula: $\text{Adjusted } R^2 = 1 - (1 - R^2) \times (n - 1) / (n - k - 1)$

Here $k = 1$ because there is one independent variable: Advertising Spend.

$$= 1 - (1 - 0.999478443) \times (15 - 1) / (15 - 1 - 1)$$

$$= 1 - (0.000521557) \times 14 / 13$$

$$= 1 - 0.000561676$$

$$= 0.999438324$$

Adjusted R Square = 0.999438324

Step 5 — Standard Error

First calculate SST (Total Sum of Squares):

$$SST = \sum(Y - \blacksquare)^2$$

$$\blacksquare = 2362/15 = 157.466667$$

$$SST = \sum(Y - 157.466667)^2 = 15969.733333$$

Next calculate SSE (Error Sum of Squares):

$$SSE = (1 - R^2) \times SST$$

$$SSE = (1 - 0.999478443) \times 15969.733333 = 8.329122$$

Degrees of freedom for error:

$$n - k - 1 = 15 - 1 - 1 = 13$$

Standard Error formula:

$$SE = \sqrt{[SSE / (n - k - 1)]}$$

$$SE = \sqrt{(8.329122 / 13)}$$

$$SE = \sqrt{(0.640701664)}$$

Standard Error = 0.80043842

Final Excel Regression Summary Output

Regression Statistics	Calculated Value
Multiple R	0.999739188
R Square	0.999478443
Adjusted R Square	0.999438324
Standard Error	0.80043842
Observations	15

Note: The calculations are shown step-by-step without using the Excel Regression shortcut, so you can understand where each Summary Output value comes from.